SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
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COMPUTER SCIENCE AND ENGINEERING

Nov – 2026

ASSIGNMENT – 1

Course Code : **CSA63**

Course Name : **THREAT INTELLIGENCE AND NETWORK SECURITY**

Max Marks**: 50**

Answer all the Questions

1	A regional hospital network stores patient records, runs an online appointment portal and uses networked diagnostic machines. In one month the hospital faces four incidents: (i) an employee e-mails a spreadsheet of patient details to the wrong recipient, (ii) ransomware encrypts the records server, (iii) a technician alters a lab report without authorisation, and (iv) a flood of requests makes the appointment portal unreachable. Apply the fundamentals of network security to this case. Explain the three goals of the CIA triad and the related goals of authentication, non-repudiation and accountability; map each incident to the goal or goals that it violates with a reason; and prepare a table that gives, for each goal, two preventive controls and one detective control that the hospital should adopt. Discuss how the hospital should balance the three goals when they conflict (for example, tight access control versus emergency access to records).	(10)	(CO1)	(BL3)
2	An online shopping company discovers that an attacker first sent a phishing e-mail to a customer-support agent, used the stolen password to log in to the internal network, moved across to the order database server, and copied customer data to an external server. Analyse the incident by classifying the types of network threats involved (malware, phishing, spoofing, sniffing, man-in-the-middle, denial of service, SQL injection and insider threats), stating which of them are passive and which are active, and identifying the likely threat actor (cybercriminal, hacktivist, nation-state or insider) with reasons. Prepare an attack-chain table that gives, for each stage, the threat, the weakness exploited, the impact on confidentiality, integrity or availability, and one control that would have stopped or detected the attack at that stage. Discuss which weaknesses were technical and which were human or procedural.	(10)	(CO1)	(BL4)
3	A security analyst at a mid-size company notices that the firewall log shows 300 failed login attempts to the VPN gateway within ten minutes from one external IP address, and that the same address appears in a public threat-sharing report as part of a credential-stuffing campaign. Using this example, explain what threat intelligence is and how it differs from raw data and information, by showing how	(10)	(CO1)	(BL3)

	the log entry is converted step by step into actionable intelligence. Describe the characteristics of good threat intelligence (timely, accurate, relevant, contextual and actionable), explain how threat intelligence changes a security team from reactive to proactive, and discuss the benefits and the limitations (for example, false positives, information overload and stale indicators) of using threat intelligence in cybersecurity.			
4	A retail chain learns that a competitor has suffered a point-of-sale malware breach. The security manager must decide how to respond and wants to present suitable threat intelligence to three groups: the Chief Information Security Officer and the board, the SOC manager, and the incident handlers and network administrators. Analyse how strategic, tactical and operational threat intelligence apply to this situation. For each type, state the question that it answers, the audience, the time horizon and the sources, and give one deliverable that you would prepare for this case (for example, a risk briefing, a report of attacker techniques and tools, a warning about an expected campaign, and a list of technical indicators). Classify the following items as strategic, tactical or operational with a reason: an annual report on ransomware trends in retail, a description of how the attacker moves from stolen vendor credentials to the POS servers, an advance warning of a planned attack on the retailer's holiday sales, and a list of malicious domains. Discuss what the company would lose if it relied on only one type.	(10)	(CO1)	(BL4)
5	A software-as-a-service company wants to set up a threat intelligence capability and begins with the requirement: "Are we likely to be targeted by credential-stuffing and ransomware groups in the next six months?" Evaluate how the threat intelligence lifecycle can be applied to answer this requirement. Describe the activities, tools and outputs of the planning and direction, collection, processing, analysis, dissemination and feedback phases, including how priority intelligence requirements are written and how information-sharing markings such as the Traffic Light Protocol are used. Then evaluate the applications of threat intelligence in the IT industry (for example, incident response, threat hunting, vulnerability and patch prioritisation, third-party risk assessment and executive risk reporting), and recommend four measures (for example, mean time to detect, mean time to respond, the percentage of alerts enriched with intelligence and the false-positive rate) to judge whether the programme is delivering value.	(10)	(CO1)	(BL5)

CSA63 – THREAT INTELLIGENCE AND NETWORK SECURITY

ASSIGNMENT – 1 | ANSWERS | NOVEMBER 2026

Q1. Apply the fundamentals of network security to the hospital case. Explain the CIA triad and related security goals.

1. Introduction

Network security means protecting computer systems, networks and data from unauthorised access, misuse, modification and attacks. A hospital stores important patient information, medical reports and appointment details. Therefore, it must maintain security to protect patient privacy and provide continuous medical services.

2. CIA Triad

The CIA triad is the basic model of network security. It consists of three main goals.

- **Confidentiality:** Keeping information secret from unauthorised people. Only authorised users can access patient records. Example: Only doctors and authorised staff can view patient medical records.
- **Integrity:** Maintaining the accuracy and correctness of information. Data should not be changed without permission. Example: A technician cannot change a patient's laboratory report without authorisation.
- **Availability:** Making information and services available whenever required. Systems should work properly without interruption. Example: The hospital appointment portal should be available 24/7.

3. Other Security Goals

Security goal	Meaning	Example
Authentication	Verifying the identity of a user	Login using password and OTP
Non-repudiation	Preventing a person from denying an action	Digital signature
Accountability	Tracking user activities	Maintaining login and access logs

4. Mapping Hospital Incidents to Security Goals

Incident	Violated goal	Reason
Patient details sent to the wrong person	Confidentiality	Private information is exposed
Ransomware encrypts records	Availability	Records become inaccessible
Technician changes a lab report	Integrity	Medical information is modified
Flood of requests makes portal unavailable	Availability	Patients cannot access appointments

5. Preventive and Detective Controls

Security goal	Two preventive controls	One detective control
Confidentiality	Strong passwords; access control	Access log monitoring
Integrity	Digital signatures; file permissions	File integrity monitoring
Availability	Regular backups; firewall protection	Network traffic monitoring

6. Balancing Security Goals

- Strict access control improves confidentiality but may delay emergency treatment.
- Hospitals should provide emergency access to authorised doctors.
- This access should be recorded and monitored.
- Regular backups help maintain availability.
- Security policies should protect patient information without affecting medical services.

Conclusion: The CIA triad helps hospitals protect patient information and network services. By using authentication, access control, backups and monitoring, hospitals can maintain confidentiality, integrity and availability.

Q2. Analyse the cyberattack on an online shopping company by classifying network threats.

1. Introduction

A network threat is any activity that can damage a computer network, steal information or interrupt services. In this case, an attacker sends a phishing email, steals an employee's password, enters the company network and copies customer data.

2. Types of Network Threats

- **Malware:** Harmful software designed to damage or access computer systems. Examples: viruses, worms and ransomware.

- **Phishing:** Sending fake emails or messages to steal personal information. Example: a fake company email asking an employee to enter their password.
- **Spoofing:** Pretending to be a trusted person, device or website. Example: an attacker creates a fake company login page.
- **Sniffing:** Secretly capturing network traffic to collect information. Example: capturing passwords sent over an unsecured network.
- **Man-in-the-Middle Attack:** An attacker secretly places themselves between two communicating parties and may read or modify communication.
- **Denial of Service (DoS):** Sending excessive requests to make a service unavailable. Example: making an online shopping website unreachable.
- **SQL Injection:** Entering malicious SQL commands into an application to access or modify database information.
- **Insider Threat:** A trusted employee or person misuses their access. Example: an employee stealing customer details.

3. Passive and Active Threats

Passive threats	Active threats
Sniffing – secretly collecting information	Malware attacks – harmful actions on systems
Monitoring network communication	Phishing – deceptive attempts to steal credentials
Usually does not directly change data	Spoofing and SQL injection
Often difficult to detect	Denial of Service – disrupts service

Note: Some attacks, such as phishing and spoofing, can involve both passive and active activities depending on how they are performed.

4. Likely Threat Actor

The most likely attacker is a **cybercriminal**. The attacker steals customer information, which may be sold or misused for financial gain. The use of stolen passwords and the copying of customer data suggest a financially motivated attack.

5. Attack Chain Table

Attack stage	Threat	Weakness exploited	Impact	Control
1. Fake email	Phishing	Lack of employee awareness	Confidentiality	Email filtering and training
2. Password stolen	Credential theft	Weak authentication	Confidentiality	Multi-factor authentication
3. Network entry	Unauthorised access	Poor access control	Confidentiality	Network access control
4. Database access	Lateral movement	Weak network separation	Confidentiality	Network segmentation
5. Data copying	Data theft	Poor data monitoring	Confidentiality	Data Loss Prevention (DLP)

6. Technical and Human Weaknesses

- **Technical:** Weak passwords, lack of multi-factor authentication, poor network segmentation, weak database access controls and lack of data monitoring.
- **Human or procedural:** Employees clicking suspicious links, lack of security awareness, poor password management and weak security policies.

Conclusion: The attack started with phishing and ended with customer data theft. Companies should use employee training, multi-factor authentication, firewalls, network segmentation and database monitoring to prevent such attacks.

Q3. Explain threat intelligence and how raw data is converted into actionable intelligence.

1. Introduction

Threat intelligence is the process of collecting, analysing and understanding information about cyber threats. It helps security teams identify attackers, understand their methods and take preventive action before serious damage occurs.

2. Difference Between Data, Information and Intelligence

Type	Meaning	Example
Data	Unprocessed facts	300 failed login attempts
Information	Organised and meaningful data	300 failed attempts from one IP in 10 minutes
Intelligence	Analysed information used for decisions	The IP is linked to credential stuffing; block it and check affected accounts

3. Conversion of Raw Data into Actionable Intelligence

A company notices 300 failed VPN login attempts from one external IP address within ten minutes. The conversion takes place in the following steps:

- **Step 1 – Data Collection:** The firewall records 300 failed login attempts. The security team collects the log details.
- **Step 2 – Processing:** The logs are organised. The IP address, time and number of attempts are identified.
- **Step 3 – Analysis:** The security team checks the IP address against public threat reports and finds that it is connected to a credential-stuffing campaign.
- **Step 4 – Interpretation:** The team understands that the attacker may be trying stolen usernames and passwords. The company identifies the possibility of account compromise.
- **Step 5 – Action:** Block the suspicious IP address, enable multi-factor authentication, review affected user accounts and monitor further login attempts.

4. Characteristics of Good Threat Intelligence

Characteristic	Explanation
Timely	Information should be received quickly.
Accurate	Information must be correct.
Relevant	It should relate to the organisation.
Contextual	It should explain the background of the threat.
Actionable	It should help the team take action.

5. Reactive Security and Proactive Security

Reactive security	Proactive security
Acts after an attack happens	Identifies threats before an attack
Finds problems after damage occurs	Monitors suspicious activities
Mainly focuses on recovery	Takes preventive actions

Example: Instead of waiting for an attacker to enter the VPN, the company identifies the suspicious IP address and blocks it in advance.

6. Benefits of Threat Intelligence

- Early detection of cyber threats.
- Helps prevent cyberattacks.
- Improves incident response.
- Protects important company information.
- Helps security teams make better decisions.
- Reduces financial losses.

7. Limitations of Threat Intelligence

- **False positives:** Normal activities may be identified as threats.
- **Information overload:** Too much information makes analysis difficult.
- **Stale indicators:** Old threat information may no longer be useful.
- **Cost:** Advanced intelligence tools can be expensive.
- **Accuracy issues:** Incorrect information may lead to wrong decisions.

Conclusion: Threat intelligence converts raw security data into useful knowledge. It helps organisations understand threats, identify suspicious activities and take preventive action to improve network security.

Q4. Analyse strategic, tactical and operational threat intelligence in the retail chain case.

1. Introduction

Threat intelligence is divided into three main types: strategic, tactical and operational. A retail company can use these types to understand and prevent point-of-sale (POS) malware attacks.

2. Types of Threat Intelligence

A. Strategic Threat Intelligence

- Provides a high-level view of cyber threats.
- Focuses on business risks and long-term security planning.

- Helps top-level management make decisions.
- **Audience:** CISO and board members. **Time horizon:** Long-term.
- **Sources:** Industry reports, security research reports, government advisories and cybersecurity trend reports.
- **Deliverable:** Annual risk briefing about ransomware and POS malware threats in the retail industry.

B. Tactical Threat Intelligence

- Explains how attackers perform their activities.
- Focuses on attacker techniques, tactics and procedures (TTPs).
- Helps security teams improve their defence methods.
- **Audience:** SOC managers and security analysts. **Time horizon:** Medium-term.
- **Sources:** MITRE ATT&CK;, threat research reports, security vendor reports and incident analysis reports.
- **Deliverable:** A report explaining how attackers use stolen vendor credentials to access POS servers.

C. Operational Threat Intelligence

- Provides details about specific attacks or campaigns.
- Helps teams prepare for and respond to immediate threats.
- **Audience:** Incident handlers and network administrators. **Time horizon:** Short-term.
- **Sources:** Threat-sharing platforms, security alerts, incident reports and threat intelligence feeds.
- **Deliverable:** A list of malicious IP addresses, domains and file hashes related to the attack.

3. Comparison Table

Feature	Strategic	Tactical	Operational
Meaning	Business-level threat view	Attacker methods	Specific attack details
Audience	CISO and board	SOC manager	Incident handlers
Time horizon	Long-term	Medium-term	Short-term
Focus	Business risks	Attack techniques	Current campaigns
Output	Risk report	TTP report	Technical indicators

4. Classification of Given Items

Given item	Type	Reason
Annual report on ransomware trends in retail	Strategic	Explains long-term industry risks
How attackers move from stolen vendor credentials to POS servers	Tactical	Explains attacker techniques
Advance warning of a planned holiday sales attack	Operational	Gives information about a specific upcoming attack
List of malicious domains	Operational	Provides technical indicators for immediate action

5. What Happens if the Company Uses Only One Type?

- **Only strategic:** The company understands business risks but may not know attack techniques or malicious indicators.
- **Only tactical:** The company understands attacker methods but may miss long-term business risks and immediate attack warnings.
- **Only operational:** The company can block known malicious indicators but may not understand the overall attacker strategy or business impact.

Conclusion: Strategic, tactical and operational intelligence work together to improve cybersecurity. Retail companies should use all three types to protect POS systems, customer information and business operations.

Q5. Evaluate the threat intelligence lifecycle and its applications in the IT industry.

1. Introduction

The threat intelligence lifecycle is a systematic process used to collect, process, analyse and share information about cyber threats. A Software-as-a-Service (SaaS) company wants to know whether it is likely to face credential-stuffing and ransomware attacks in the next six months. The threat intelligence lifecycle helps the company answer this question.

2. Threat Intelligence Lifecycle



Feedback helps improve the next intelligence cycle.

3. Phases of the Threat Intelligence Lifecycle

1. Planning and Direction

- The company identifies its security requirements and decides what information it needs.
- It defines Priority Intelligence Requirements (PIRs). A PIR is an important question that the security team needs to answer.
- **Example PIRs:** Are we likely to be targeted by credential-stuffing groups? Are ransomware groups targeting SaaS companies? What are the possible threats in the next six months?
- **Output:** Intelligence requirements and collection plan.

2. Collection

- The company collects information from different sources about attackers and their activities.
- It gathers firewall logs, login records and security reports.
- **Tools and sources:** Threat intelligence feeds, firewall logs, SIEM tools, security vendor reports and government cybersecurity advisories.
- **Output:** Raw threat data.

3. Processing

- Collected data is organised and filtered.
- Duplicate and unnecessary information is removed and data is converted into a useful format.
- Suspicious IP addresses and domains are identified.
- **Tools:** SIEM platforms and data-processing tools. **Output:** Clean and organised threat information.

4. Analysis

- Security analysts study the processed information.
- They identify possible attackers and attack methods and estimate the risk to the company.
- They predict possible credential-stuffing and ransomware attacks.
- **Output:** Analysed threat intelligence and risk reports.

5. Dissemination

- The analysed intelligence is shared with the correct teams.
- Technical information is sent to security analysts, business risk reports to management and urgent warnings to incident response teams.
- **Output:** Security alerts, reports and threat warnings.

6. Feedback

- The security team collects feedback from users and managers.
- It checks whether the intelligence was useful and identifies mistakes or missing information.
- The collection and analysis process is improved.
- **Output:** Improved intelligence requirements and processes.

4. Traffic Light Protocol (TLP)

TLP is used to control how threat intelligence information can be shared.

TLP marking	Meaning
TLP:RED	Only specific recipients can access the information.
TLP:AMBER	Limited sharing with authorised people.
TLP:GREEN	Sharing within a trusted community.
TLP:CLEAR	Information can be shared publicly.

It helps organisations share threat information safely without exposing sensitive details.

5. Applications of Threat Intelligence in the IT Industry

- **Incident Response:** Helps identify and understand cyberattacks and supports quick recovery.
- **Threat Hunting:** Helps security teams search for hidden threats and identify suspicious network activities.
- **Vulnerability and Patch Prioritisation:** Helps identify dangerous vulnerabilities and gives priority to important security updates.
- **Third-Party Risk Assessment:** Helps check the security risks of suppliers and service providers.
- **Executive Risk Reporting:** Provides information about cybersecurity risks and helps management make security decisions.

6. Measures to Evaluate Threat Intelligence

Measure	Explanation
Mean Time to Detect (MTTD)	Average time taken to detect a threat.
Mean Time to Respond (MTTR)	Average time taken to respond to an incident.
Intelligence-enriched alerts	Percentage of security alerts improved with threat intelligence.
False-positive rate	Percentage of normal activities incorrectly identified as threats.

Conclusion: The threat intelligence lifecycle helps organisations identify, analyse and prevent cyber threats. By following all six phases and measuring performance, a SaaS company can improve its security and reduce the risk of credential-stuffing and ransomware attacks.